

Topic 8: Redox Chemistry and Groups 1, 2, and 7 This

comprehensive guide covers every specification point in **Topic 8** for Edexcel IAL Chemistry Unit 2, broken down into **8A (Redox Chemistry)**, **8B (Groups 1 and 2)**, and **8C (Group 7)**, including all equations, colors, observations, and exam-assessed practical details 1. --- **### PART 8A:**

REDOX CHEMISTRY ##### Specification Points 8.1 & 8.2: Oxidation Numbers and Rules

An **oxidation number** (or oxidation state) represents the formal charge an atom would have if all bonds in a compound were purely ionic 1. **##### Rules for Assigning Oxidation**

Numbers: 1. **Uncombined Elements:** Any element in its uncombined elemental state has an oxidation number of **0** (e.g. H_2 , O_2 , Fe , $\text{S}_8 = 0$) 1. 2. **Simple Monatomic Ions:**

Equal to the charge on the ion (e.g. $\text{Na}^+ = +1$, $\text{Mg}^{2+} = +2$, $\text{Fe}^{3+} = +3$, $\text{Cl}^- = -1$) 1. 3. **Group 1, 2, and 3 Metals in Compounds:** Always **+1**, **+2**, and **+3** respectively 1, 2. 4. **Fluorine in Compounds:** Always **-1** (most electronegative element) 1, 2. 5. **Hydrogen in Compounds:** Always **+1**, **except in metal hydrides** (e.g. NaH , CaH_2) where hydrogen exists as a hydride ion (H^-) with an oxidation number of **-1** 1, 2. 6. **Oxygen in Compounds:** Always **-2**, with three notable exceptions: * **Peroxides** (e.g. H_2O_2 , Na_2O_2): Oxygen is **-1** 1, 2. * **Superoxides** (e.g. KO_2): Oxygen is $-\frac{1}{2}$ 2. * **Compounds with Fluorine** (e.g. F_2O) where $\text{O} = +2$, and O_2F_2 where $\text{O} = +1$) 1, 2. 7. **Neutral Molecules:** The sum of oxidation numbers of all atoms in a neutral molecule equals **0** 1, 2. 8. **Polyatomic Ions:** The sum of oxidation numbers equals the **overall charge on the ion** (e.g. in SO_4^{2-} , $\text{S} = +6$ and $4 \times (-2) = -8$, sum $= -2$) 1, 2. --- **##### Specification Points 8.3 & 8.4: Stock Notation and Chemical Formulae**

* **Stock Notation:** Roman numerals in parentheses indicate the positive oxidation state of an element that exhibits variable oxidation numbers 1, 3. * FeCl_2 : Iron(II) chloride vs FeCl_3 : Iron(III) chloride 1, 3. *

K_2SO_3 : Potassium sulfate(IV) vs

K_2SO_4 : Potassium sulfate(VI) 1. * NaClO :

Sodium chlorate(I) vs NaClO_3 : Sodium chlorate(V) 1. * NaNO_2 : Sodium nitrate(III) vs NaNO_3 : Sodium nitrate(V) 1. --- ##### **Specification Points 8.5, 8.6 & 8.9: Oxidation, Reduction, and Redox Agents** | Process | Electron Transfer | Oxidation Number Change | Oxygen / Hydrogen | Behavior of Elements | | :--- | :--- | :--- | :--- | :--- | | **Oxidation** | **Loss** of electrons (OIL) 4 | **Increase** in oxidation number 5, 6 | Gain of Oxygen / Loss of Hydrogen 4 | Metals (s-block) lose electrons to form positive cations 5, 7. | | **Reduction** | **Gain** of electrons (RIG) 4 | **Decrease** in oxidation number 5, 6 | Loss of Oxygen / Gain of Hydrogen 4 | Non-metals (p-block) gain electrons to form negative anions 5, 7. | * **Oxidising Agent**: A species that **gains electrons** (is reduced) and causes another species to be oxidised. Its oxidation number **decreases** 5, 8, 9. * **Reducing Agent**: A species that **donates electrons** (is oxidised) and causes another species to be reduced. Its oxidation number **increases** 5, 9. --- ##### **Specification Points 8.7 & 8.8: Disproportionation Reactions** A **disproportionation reaction** is a specific redox reaction where **an element in a single species is simultaneously oxidised and reduced** 5, 10, 11. ##### **Key Examples**: 1. **Chlorine in Water**: $\text{Cl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HCl}(\text{aq}) + \text{HClO}(\text{aq})$ * Chlorine's oxidation state changes from **0** in Cl_2 to **-1** in HCl (reduction) and **+1** in HClO (oxidation) 5, 10. 2. **Decomposition of Hydrogen Peroxide**: $2\text{H}_2\text{O}_2(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ * Oxygen changes from **-1** in H_2O_2 to **-2** in H_2O (reduction) and **0** in O_2 (oxidation) 12, 13. --- ##### **Specification Point 8.10: Ionic Half-Equations and Full Equations** Half-equations display explicit electron transfers for individual oxidation and reduction processes 14. ##### **Constructing Half-Equations in Acidic Conditions**: 1. Balance atoms being oxidised/reduced 15, 16. 2. Add H_2O to balance oxygen atoms 15, 16. 3. Add H^+ ions to balance hydrogen atoms 15, 16. 4. Add e^- to balance overall electrical charges 15, 16. ##### **Example**: Combining Fe^{2+} oxidation and $\text{Cr}_2\text{O}_7^{2-}$ reduction: * **Oxidation**: $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$ (multiply by 6) 17, 18 *

Reduction: $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ 19 * **Overall**

Equation: $6\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ \rightarrow 6\text{Fe}^{3+} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ 18 ---

PART 8B: THE ELEMENTS OF GROUPS 1 AND 2

Specification Point 8.11: Trend in Ionisation Energy * **Trend:** First and second ionisation energies **decrease down Groups 1 and 2** 20, 21.

* **Explanation:** 1. Atomic radius increases down the group 21. 2. The number of inner filled electron shells increases, leading to **increased shielding** 21. 3. Increased distance and shielding outweigh the increase in nuclear charge, weakening the electrostatic attraction between the nucleus and outermost valence electron(s) 21. --- ##### **Specification**

Point 8.12: Trend in Reactivity * **Trend:** Chemical reactivity **increases down Groups 1 and 2** 20, 21. * **Explanation:** Groups 1 and 2 metals

react by losing outer valence electrons to form positive ions

(M^+ or M^{2+}) 20. Because ionisation energies decrease down the group, less energy is required to remove valence electrons, facilitating faster and more vigorous reactions 21. --- #####

Specification Point 8.13: Reactions with Oxygen, Chlorine, and Water ##### **1. Reaction with Oxygen (O_2):** * **Group 1:**

$4\text{M}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{M}_2\text{O}(\text{s})$ (White basic solid oxides) 22, 23. * **Group 2:**

$2\text{M}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MO}(\text{s})$ (White basic solid oxides) 22, 23. * **Special Notes:**

* Lithium forms simple oxide Li_2O ; Sodium forms peroxide Na_2O_2 ; Potassium forms superoxide KO_2 2. * Group 2 metals burn with vivid flame colors (e.g.

Mg) burns with a blinding bright white flame to form

MgO 24. ##### **2. Reaction with Chlorine (Cl_2):**

* **Group 1:** $2\text{M}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{MCl}(\text{s})$ (White ionic chlorides) 23, 25. * **Group 2:**

$\text{M}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow \text{MCl}_2(\text{s})$ (White ionic chlorides) 23, 26. ##### **3. Reaction**

with Water (H_2O): * **Group 1:** $2\text{M}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{MOH}(\text{aq}) + \text{H}_2(\text{g})$ 23. * **Observations:** Effervescence, metal melts into

a sphere (Na, K), floats, moves rapidly across surface, disappears;

solution turns alkaline 24. Potassium catches fire with a lilac flame 24. *

Group 2: $M(s) + 2H_2O(l) \rightarrow M(OH)_2(aq/s) + H_2(g)$ 24. *

Magnesium anomaly with Water vs Steam: * With **cold water**,

$Mg(s)$ reacts extremely slowly: $Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(s) + H_2(g)$ 24 * With **steam** ($H_2O(g)$), $Mg(s)$ reacts vigorously, burning with a bright white flame to form

solid Magnesium Oxide: $Mg(s) +$

$H_2O(g) \rightarrow MgO(s) +$

$H_2(g)$ 24, 27 --- ##### **Specification Point 8.14:**

Reactions of Oxides and Hydroxides ##### 1. Reaction of Oxides

with Water: * Group 1: $M_2O(s) +$

$H_2O(l) \rightarrow 2MOH(aq)$ 28, 29 *

Group 2: $MO(s) + H_2O(l) \rightarrow$

$M(OH)_2(aq)$ 29 * *Ionic Equation:*

$O^{2-}(s) + H_2O(l) \rightarrow$

$2OH^-(aq)$ 28, 30 ##### **2. Neutralisation Reactions with**

Dilute Acids: * **Oxides:** $MO(s) + 2HCl(aq)$

$\rightarrow MCl_2(aq) + H_2O(l)$ 31 *

Hydroxides: $M(OH)_2(s) + 2HCl(aq)$

$\rightarrow MCl_2(aq) + 2H_2O(l)$ 32 *

Crucial Exam Note on H_2SO_4 : Reacting

$Ca(OH)_2$, $Sr(OH)_2$ or

$Ba(OH)_2$ with dilute H_2SO_4 forms an

insoluble sulfate layer ($BaSO_4$) over the solid, preventing

further reaction 32, 33. --- ##### **Specification Point 8.15: Trends in**

Solubility of Group 2 Hydroxides and Sulfates Group 2 Hydroxides

$[M(OH)_2]$ Group 2 Sulfates $[MSO_4]$ $Mg(OH)_2$ (Insoluble) $MgSO_4$

(Soluble) | | Solubility | Solubility | INCREASES | DECREASES ▼

▼ $Ba(OH)_2$ (Soluble) $BaSO_4$ (Insoluble White Ppt) * **Hydroxide**

Solubility: Increases down Group 2 ($Mg(OH)_2$)

insoluble; $Ca(OH)_2$ sparingly soluble;

$Ba(OH)_2$ very soluble) 33, 34. As solubility increases,

OH^- concentration increases, making solutions **more alkaline**

down the group 33. * **Sulfate Solubility: Decreases down Group 2**

($MgSO_4$) soluble; ($BaSO_4$) completely insoluble

white precipitate) 33, 34. * (Exam Tip: All Group 1 hydroxides and sulfates are completely soluble in water) 35. --- #####

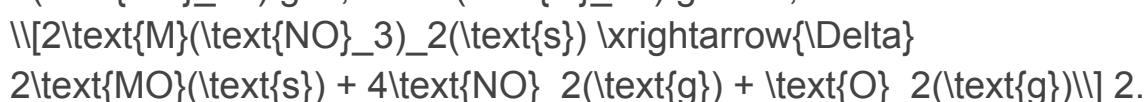
Specification Point 8.16: Thermal Stability of Nitrates and Carbonates *

Definition: Thermal stability measures a compound's resistance to decomposition upon heating 34, 36. * **Trend:** Thermal stability of Group 1 and 2 nitrates and carbonates **increases down the group** 37, 38.

Mechanism (Polarising Power of Cation): 1. Down the group, cation charge remains $+1$ or $+2$, but **cation ionic radius increases** 39, 40. 2. **Charge density** of the cation decreases down the group 39, 40. 3. Smaller cations with high charge density (e.g. Li^+ , Mg^{2+}) exert a strong **polarising power** on large complex anions (CO_3^{2-} or NO_3^-), distorting their electron cloud 39, 41, 42. 4. This weakens the internal covalent $\text{C}-\text{O}$ or $\text{N}-\text{O}$ bonds within the anion, lowering the thermal energy required to decompose the salt 38, 39, 41. Small Cation (High Charge Density) $\longrightarrow$ Polarises Anion Electron Cloud $\longrightarrow$ Weakens C-O/N-O Bond $\longrightarrow$ Low Thermal Stability

Thermal Decomposition Patterns: Group 2 Nitrates & LiNO_3 : $2\text{M}(\text{NO}_3)_2 \longrightarrow 2\text{MO} + 4\text{NO}_2$ (Brown Fumes) + O_2 Group 1 Nitrates (Na to Cs): $2\text{MNO}_3 \longrightarrow 2\text{MNO}_2 + \text{O}_2$ (No Brown Fumes) Group 2 Carbonates & Li_2CO_3 : $\text{MCO}_3 \longrightarrow \text{MO} + \text{CO}_2$ Group 1 Carbonates (Na to Cs): Thermally stable at Bunsen temperatures (No Decomposition) 1. **Group 2**

Nitrates: All decompose on heating to form metal oxide, brown toxic NO_2 gas, and O_2 gas 37, 43:



Group 1 Nitrates: Lesser decomposition for Na to Cs (forms metal nitrite and O_2 ; no brown fumes) 43-45: $2\text{MNO}_3 \xrightarrow{\Delta} 2\text{MNO}_2 + \text{O}_2$

* **Lithium Anomaly:**

Li^+ is small enough to polarise NO_3^- , so LiNO_3 decomposes like a Group 2 nitrate 43, 44: $4\text{LiNO}_3 \xrightarrow{\Delta} 2\text{Li}_2\text{O} + 4\text{NO}_2 + \text{O}_2$

3. **Group 2 Carbonates:** All decompose on heating to metal oxide and CO_2 gas 40, 44: $\text{MCO}_3 \xrightarrow{\Delta} \text{MO} + \text{CO}_2$ 4. **Group 1 Carbonates:** Thermally stable at

Bunsen temperatures (do not decompose), **except**

Li_2CO_3 42, 46, 47: $\text{Li}_2\text{CO}_3(\text{s}) \xrightarrow{\Delta} \text{Li}_2\text{O}(\text{s}) + \text{CO}_2(\text{g})$

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* **Mechanism:** Heat energy from the flame promotes electrons from ground state to higher quantum energy levels (excited state) 48. As electrons drop back to lower levels, energy is emitted as visible light of specific wavelength ($\Delta E = hf$) 48. | Cation | Flame Color |
Cation | Flame Color | :--- | :--- | :--- | :--- | | Li^+ | Red / Crimson 49 | Be^{2+} | **No color** (energy gap outside visible spectrum) 49, 50 | | Na^+ | Yellow / Orange 49 |
 Mg^{2+} | **No color** (energy gap outside visible spectrum) 49, 50 | | K^+ | Lilac 49 | Ca^{2+} | Brick red 49 | |
 Rb^+ | Red / Purple 49 | Sr^{2+} | Crimson red 49 | |
 Cs^+ | Blue / Violet 49 | Ba^{2+} | Apple green 49

Flame Test Procedure: 1. Dip a clean platinum or nichrome wire into **concentrated HCl** 51. 2. Heat wire in a non-luminous Bunsen flame to clean it until no color is imparted. (Conc. HCl converts compounds into volatile chloride salts) 51. 3. Dip wire in conc. HCl, pick up sample, place in flame, and observe color 51. --- ##### **Specification Point 8.18(i) & 8.19: Qualitative Tests for Anions & Cations**

1. **Carbonate** (CO_3^{2-}) and **Hydrogencarbonate** (HCO_3^-): * Add dilute acid (HCl) or HNO_3 . Effervescence produced 52, 53. * Bubble gas through **limewater** ($\text{Ca}(\text{OH})_2$); turns milky due to white precipitate of CaCO_3 41, 52, 53. *

$\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ 53

2. **Sulfate** (SO_4^{2-}): * Add **dilute HCl** or HNO_3 (to destroy any CO_3^{2-} ions that would form false precipitates), then add **aqueous BaCl₂** or

$\text{Ba}(\text{NO}_3)_2$ 51, 53. * Formation of an **insoluble white precipitate of BaSO₄** 53. * $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ 53

3. **Ammonium** (NH_4^+): * Add aqueous NaOH and warm gently 53. * Pungent ammonia gas (NH_3) turns **damp red litmus paper blue**, or forms dense white fumes of

NH_4Cl when tested with glass rod dipped in conc.

HCl 53. $\text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

$\rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$ 53 --- ###

PART 8C: INORGANIC CHEMISTRY OF GROUP 7 (HALOGENS) ####

Specification Point 8.24: Trends in Group 7 Elements | Halogen |

Formula | Physical State at RTP | Color (Standard) | Color in Water |

Color in Cyclohexane (Organic) | | :--- | :--- | :--- | :--- | :--- | :--- | |

Fluorine | F_2 | Gas | Pale Yellow 54 | Reacts violently |

Reacts violently | | **Chlorine** | Cl_2 | Gas | Yellow-Green 54 |

Pale Green 55 | Pale Green 55 | | **Bromine** | Br_2 | Liquid |

Red-Brown 54 | Orange / Yellow 56 | **Orange / Red** 56 | | **Iodine** |

I_2 | Solid | Dark Grey / Black 54 | Brown 55, 56 | **Violet /**

Purple 55, 56 | | **Astatine** | At_2 | Solid | Black 55 | — | — |

Explanations of Periodic Trends: 1. Melting & Boiling Points

(Increase Down Group): Halogens exist as non-polar diatomic

molecules held by **London dispersion forces** 55, 57. As molecular size

and total electron count increase ($\text{F}_2: 18\text{e}^- \rightarrow$

$\text{I}_2: 106\text{e}^-$), electron cloud fluctuations increase,

strengthening London forces and requiring higher thermal energy to

separate molecules 55, 57. 2. **Electronegativity (Decreases Down**

Group): Atomic radius and inner electron shielding increase down the

group, weakening the electrostatic pull of the nucleus on shared bonding

electrons 58, 59. 3. **Reactivity & Oxidising Power (Decreases Down**

Group): Halogens react by gaining an electron to form $(1-)$ halide ions

($\text{X}_2 + 2\text{e}^- \rightarrow 2\text{X}^-$) 58. Down the group,

outer shells are further from the nucleus with greater shielding, making it

harder to attract an incoming electron 58, 59. --- ##### **Specification**

Point 8.25: Halogen Displacement Reactions A more reactive halogen

displaces a less reactive halide ion from aqueous solution 60, 61: *

$\text{Cl}_2(\text{aq}) + 2\text{Br}^-(\text{aq}) \rightarrow$

$2\text{Cl}^-(\text{aq}) + \text{Br}_2(\text{aq})$ (Solution turns

yellow/orange) 62, 63 * $\text{Cl}_2(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow$

$2\text{Cl}^-(\text{aq}) + \text{I}_2(\text{aq})$ (Solution turns

brown) 62, 63 * $\text{Br}_2(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow$

$2\text{Br}^-(\text{aq}) + \text{I}_2(\text{aq})$ (Solution turns brown) 64

Adding Cyclohexane (Organic Layer): Halogens are non-polar

and much more soluble in cyclohexane than in water 56. Adding

cyclohexane and shaking separates the mixture into two layers: top organic layer shows distinct colors (**Orange for Br_2** ,

Purple/Violet for I_2) 56, 61. --- ##### **Specification Point**

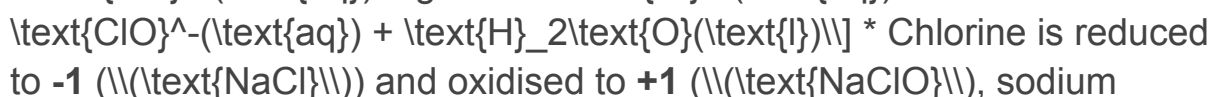
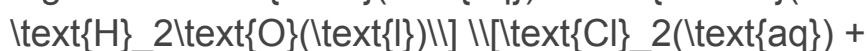
8.26: Disproportionation Reactions of Chlorine

1. Water Treatment (Sterilisation): $\text{Cl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l})$



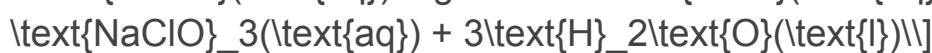
* Chlorine is reduced to **-1** in HCl and oxidised to **+1** in chloric(I) acid (HClO) 10, 65. HClO acts as a sterilising agent killing bacteria 66, 67.

2. With COLD Dilute Alkali (15°C) / (20°C): $\text{Cl}_2(\text{aq}) + 2\text{NaOH}(\text{aq})$



* Chlorine is reduced to **-1** (NaCl) and oxidised to **+1** (NaClO), sodium chlorate(I) / active ingredient in household bleach) 67, 68.

3. With HOT Concentrated Alkali (70°C): $3\text{Cl}_2(\text{aq}) +$



* Chlorine is reduced to **-1** (NaCl) and oxidised to **+5** (NaClO_3), sodium chlorate(V) / weedkiller) 69, 70. --- ##### **Specification Point 8.27(i): Reactions of Solid**

Halides with Concentrated Sulfuric Acid * Trend in Halide Reducing

Ability: Increases down Group 7 ($\text{F}^- < \text{Cl}^- < \text{Br}^-$

$< \text{I}^-$) 71, 72. Larger halide ions have lower electron attraction

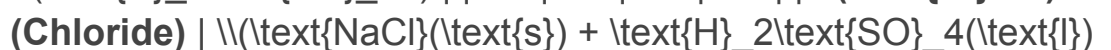
due to increased radius and shielding, donating electrons more easily

($2\text{X}^- \rightarrow \text{X}_2 + 2\text{e}^-$) 58, 59. | Halide Ion |

Reaction Equations | Observations & Products | Role of Conc.

H_2SO_4 | | :--- | :--- | :--- | :--- | | Cl^-

(Chloride) | $\text{NaCl}(\text{s}) + \text{H}_2\text{SO}_4(\text{l})$



73, 74 | Steamy white fumes of $\text{HCl}(\text{g})$ 73, 74. | **Acid ONLY**

(Cl^-) cannot reduce H_2SO_4) 71. | |

Br^- **(Bromide)** | 1. $\text{NaBr} + \text{H}_2\text{SO}_4$



73, 74

- $$2\text{HBr} + \text{H}_2\text{SO}_4 \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$$
 74 | Red-brown gas of Br_2 , choking gas of SO_2 , steamy fumes of HBr 74. | **Oxidising Agent** (S) reduced from +6 to +4 71, 74. | | **Iodide** (I^-) 1. $\text{NaI} + \text{H}_2\text{SO}_4 \rightarrow \text{HI} + \text{NaHSO}_4$ 74
 - $$2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$$
 ($\text{S} = +4$) 74
 - $$6\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 3\text{I}_2 + \text{S} + 4\text{H}_2\text{O}$$
 ($\text{S} = 0$) 74
 - $$8\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$$
 ($\text{S} = -2$) 74 | Purple vapour/dark solid of I_2 , yellow solid of S , **bad egg smell of H_2S** , choking SO_2 74. | **Strong Oxidising Agent** (S) reduced from +6 to +4, 0, and -2 71, 74. | --- **Specification Point 8.27(ii): Silver Nitrate and Ammonia Tests for Halide Ions** **Procedure:** 1. Acidify test solution with **dilute nitric acid** (HNO_3) to prevent false precipitates from CO_3^{2-} or OH^- ions 75. 2. Add **silver nitrate solution** (AgNO_3) 75. 3. Add **dilute aqueous ammonia** (NH_3), followed by **concentrated aqueous NH_3** 75. Unknown Halide $\rightarrow$ Add Dilute HNO_3 + AgNO_3 |

			▼	▼	▼	White Ppt	Cream Ppt	Yellow
Ppt (AgCl)	(AgBr)	(AgI)				Soluble in Dilute NH_3	Dilute NH_3 , both	Conc NH_3
						Soluble in Dilute NH_3	Conc NH_3	* Chloride (Cl^-): White precipitate (AgCl) $\rightarrow$ Soluble in dilute NH_3 72, 75. $[\text{AgCl}(\text{s}) + 2\text{NH}_3(\text{aq}) \rightarrow \text{Ag}(\text{NH}_3)_2^+(\text{aq}) + \text{Cl}^-(\text{aq})]$ 75
						Soluble in Dilute NH_3	Conc NH_3	* Bromide (Br^-): Cream precipitate (AgBr) $\rightarrow$ Insoluble in dilute NH_3 , soluble in concentrated NH_3 72, 75.
						Soluble in Dilute NH_3	Conc NH_3	* Iodide (I^-): Yellow precipitate (AgI) $\rightarrow$ Insoluble in both dilute and concentrated NH_3 72, 75. --- Specification Point 8.27(iii): Reactions of Hydrogen Halides
1. **With Ammonia Gas** (NH_3): Forms dense white smoke of ammonium halide 76, 77:

$$\text{HX}(\text{g}) + \text{NH}_3(\text{g}) \rightarrow \text{NH}_4\text{X}(\text{s})$$
 76, 77 2. **With Water:** Dissolves readily in water to form strong acidic solutions 76, 78:
$$\text{HCl}(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_3\text{O}^+(\text{aq}) + \text{Cl}^-(\text{aq})$$
 76, 78 --- **Specification Point 8.28: Predictions for Fluorine and Astatine** * **Fluorine** (F_2): Most reactive halogen, highest electronegativity, strongest oxidising agent 59. * **Critical Exam Exception:** **Silver fluoride** (AgF) is **SOLUBLE in water**, so the acidified silver nitrate test cannot be used to detect F^- ions 76, 79! * **Astatine** (At_2): Dark black solid, highest melting point, lowest electronegativity, weakest oxidising agent, At^- is the strongest reducing agent 55, 71, 72. --- 🧠 **Next Step:** Would you like to practice a set of exam-style questions on **Topic 8**, or are you ready to move on to **Topic 9 / Topic 10 (Kinetics, Equilibria & Organic Chemistry)** next?